# Jon Wick

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## Summary

Molecular and computational biologist with experience in Western blots, PCR, and sterile cell culture, as well as genomic analysis. Experience in investigating RNA interference pathways in *T. thermophila* and protein degradation pathways in *C. elegans*. I am seeking a laboratory technician position to apply these molecular and computational skills to immunology and cancer research.

## Education

**B.S. Molecular and Cellular Biology**, Computer Science minor — September 2023-June 2026
Western Washington University – Bellingham, WA

- **GPA**: 3.5/4.0
- **Honors:** Outstanding Graduate, WWU Dept of Biology, 2026
- **Relevant Upper Division Coursework:** Immunobiology, Advanced Cell and Molec Lab, Biochemistry, Genomic Data Analysis, Cellular and Molecular Biology, Microbiology, Biostatistics, Genetics, Organic Chemistry.

**Tacoma Community College** – Dual Enrollment (High School) — September 2021-June 2023

## Skills and Technologies

**Wet Lab:** Western Blot, PCR, Reverse Transcriptase PCR, Cell Culture, Aseptic Technique

**Bioinformatics:** Genomic Data Analysis, DESeq2, bedtools, samtools, Data Visualization (ggplot2, matplotlib)

**Programming:** Python, R, Java, C++, C, SQL, Bash

**Software:** Benchling, Command Line Interface, Git, ImageJ, Jupyter Notebook, Linux, Microsoft Office/Libre Office, NeoVim, SnapGene, SSH

## Research Experience

**Undergraduate Researcher**, Lee Lab, WWU – Bellingham, WA — June 2024 - present

- Investigated roles of small non-coding RNAs in an RNAi pathway maintaining genome integrity in *Tetrahymena thermophila*, combining computational biology with molecular biology techniques (Western Blot, cell culture, aseptic technique).
- Developed custom bioinformatics pipelines in Python and R for genome-wide sRNA biogenesis loci identification and differential expression analysis.
- Maintained live cell cultures under sterile conditions and performed protein analysis via Western Blot to characterize RNAi pathway components.
- Presented findings at lab meetings, journal clubs, and the WWU SURP poster session.
- Prepared media, buffers, reagents, and performed lab duties for daily lab operations.

**Advanced Molecular and Cell w/lab** – BIOL 487, WWU — Spring 2025

- Investigated ER-associated protein degradation (ERAD) via E2 ubiquitin ligase *ubc-6* in *C. elegans*, a pathway directly relevant to protein quality control in cancer and neurodegeneration.
- Performed Western Blots, Reverse Transcriptase PCR, and endpoint PCR to quantify misfolded protein accumulation in *ubc-6* knockout lines.
- Co-authored a poster presented at the Northwest Worm Meeting and co-wrote a micropublication on research findings.
- Maintained detailed electronic lab notebooks using Benchling.

**Genomic Data Analysis** – BIOL 477, WWU — Winter 2025

- Conducted mRNA differential expression analysis on *Tetrahymena thermophila* datasets using DESeq2, identifying differentially expressed genes across experimental conditions.
- Processed and analyzed genomic data in a Unix environment using Bash, Python, and high-performance computing resources.
- Communicated findings through a written research paper and oral presentation.

**Biotech Research Intern**, Rain Incubator – Tacoma, WA — May 2022 - December 2022

- Designed and executed pilot PCR-based experiments for tracking European Green Crab migration.
- Gained foundational experience including hypothesis development, literature review, and experimental planning.

## Publications

**Understanding the Leadership Structure and Mentoring Model of an Extracurricular Robotics Team: Key Findings from a Case Study** March 2024

*SIGCSE 2024: Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2*
Hitender Oswal, ***Jon Wick***, Seth Tandon, Ashley Brewster, Sushil K Oswal

DOI: 10.1145/3626253.3635603

## Posters

**Re-identification of the genomic origins of small RNAs in the model organism *Tetrahymena thermophila* and characterization of those dependent on genome protective *RDN2***
*2025 SURP poster session*, October 2025.
**Jon Wick**, Noah Haight, Hannah Thorp, Dan Pollard, and Suzanne R. Lee

**The Role of E2 Ligases ubc-6 and 7 in ERAD in C. elegans**
*Northwest Worm Meeting*, June 2025.
Summer Simpson, Abby Mellin, Dylan Wilding, Natalie Patch, **Jon Wick**, Bree Nelson, Hannah Thorp, Caelin Sickler, and Lina Dahlberg

**Remapping previously sequenced 23-24 nucleotide sRNAs to the 2020 Tetrahymena thermophila Genome**
*WWU SURP poster session*, September 2024.
**Jon Wick**, Noah Haight, Suzanne R. Lee, Dan Pollard

**Leadership Structure and Mentoring Model of an Extracurricular Robotics Team**
*ACM Special Interest Group Computer Science Education conference*, March 2024.
Hitender Oswal, **Jon Wick**, Seth Tandon, Ashley Brewster, Sushil K. Oswal

## Awards

**Declan Barron Memorial Scholarship** 2025
Western Washington University, Dept of Biology
**WWU Biology Summer Undergraduate Research Program award** 2025
Western Washington University, Dept of Biology

## Additional Experience

**Club Lead**, Molecular Bioscience Club, WWU – Bellingham, WA August 2025 - June 2026

- Organize events to build community and provide information about programs, research opportunities, and professional development to students of all course standings.
- Assisted students in getting into research labs.
- Collaborated with faculty within and outside of WWU on events.

**Computing Education Research**, Sushil K. Oswal, UW – Tacoma, WA February 2023 - May 2024

- Co-authored and published a paper at the ACM SIGCSE conference on mentorship structures in STEM education.
- Performed qualitative analysis on interviews and survey responses.

**Programming Lead**, SOTAbots FIRST Team 2557 – Tacoma, WA September 2019 - June 2023

- Developed Code used in the semi-finals of the 2022 World Championship of FIRST Robotics Competition and 2023 district championship.
- Taught students to program in Java using industry-standard design patterns.
- Coordinated with other leads on the team to develop a robot, build community, perform local outreach, and obtain funding.